



C#

Level 1

Unary Operators



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Definition

 Unary Operators

 Operators that work on only one operand.

Operators

 $+x$

- ✓ Unary Plus
- ✓ Keeps the value unchanged.

 $-x$

- ✓ Unary Minus
- ✓ Changes the sign.

 $++x$ / $x++$

- ✓ Increment
- ✓ Increases the value by 1.

 $--x$ / $x--$

- ✓ Decrement
- ✓ Decreases the value by 1.

 $!x$

- ✓ Logical NOT
- ✓ Reverses a Boolean value.
- ✓ $\text{true} \rightarrow \text{false}$
- ✓ $\text{false} \rightarrow \text{true}$

💡 Example

```
result = +number;  
Console.WriteLine("+number = " + result);  
result = -number;  
Console.WriteLine("-number = " + result);  
result = ++number;  
Console.WriteLine("++number = " + result);  
result = --number;  
Console.WriteLine("--number = " + result);  
Console.WriteLine("!flag = " + (!flag));  
Console.WriteLine(("number++ = ", number++));  
Console.WriteLine(("number = ", number));  
Console.WriteLine(("++number = ", ++number));  
Console.WriteLine(("number = ", number));
```

➡ Result

```
+number = 10  
-number = -10  
++number = 11  
--number = 10  
!flag = False  
(number++ = , 10)  
(number = , 11)
```

Note

- ◆ `++x` and `x++` both increase by 1, but differ in when the value is used.
- ◆ `--x` and `x--` both decrease by 1, but differ in when the value is used.



Comparison

Prefix

 $++x \rightarrow$ Increase first, then use the value.

 $--x \rightarrow$ Decrease first, then use the value.

Postfix

 $x++ \rightarrow$ Use the current value, then increase.

 $x-- \rightarrow$ Use the current value, then decrease.

✦ Summary

- ✓ Unary operators work on one operand.
- ✓ `+` → Keeps the value.
- ✓ `-` → Changes the sign.
- ✓ `++` → Increase by 1.
- ✓ `--` → Decrease by 1.
- ✓ `!` → Reverses Boolean value.

Thank you 😊